2.3.2 Physical properties:

Some of the physical properties of the group 14 elements are listed below

Table 2.4 Physical properties of group 14 elements

Property	Carbon	Silicon	Germanium	Tin	Lead
Physical state at 293 K	Solid	Solid	Solid	Solid	Solid
Atomic Number	6	14	32	50	82
Isotopes	^{12}C, ^{13}C, ^{14}C	^{28}Si, ^{30}Si	^{73}Ge, ^{74}Ge	^{120}Sn	^{208}Pb
Atomic Mass ($g.mol^{-1}$ at 293 K)	12.01	28.09	72.63	118.71	207.2
Electronic configuration	$[He]2s^2\ 2p^2$	$[Ne]3s^2\ 3p^2$	$[Ar]3d^{10}\ 4s^2\ 4p^2$	$[Kr]4d^{10}\ 5s^2\ 5p^2$	$[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^2$
Atomic radius (Å)	1.70	2.10	2.11	2.17	2.02
Density ($g.cm^{-3}$ at 293 K)	3.51	2.33	5.32	7.29	11.30
Melting point (K)	Sublimes at 4098	1687	1211	505	601
Boiling point (K)		3538	3106	2859	2022

2.3.3 Tendency for catenation

Catenation is an ability of an element to form chain of atoms. The following conditions are necessary for catenation. (i) the valency of element is greater than or equal to two, (ii) element should have an ability to bond with itself (iii) the self bond must be as strong as its bond with other elements (iv) kinetic inertness of catenated compound towards other molecules. Carbon possesses all the above properties and forms a wide range of compounds with itself and with other elements such as H, O, N, S and halogens.

2.3.4 Allotropes of carbon

Carbon exists in many allotropic forms. Graphite and diamond are the most common allotropes. Other important allotropes are graphene, fullerenes and carbon nanotubes.

Graphite is the most stable allotropic form of carbon at normal temperature and pressure. It is soft and conducts electricity. It is composed of flat two dimensional sheets of carbon atoms. Each sheet is a hexagonal net of sp^2 hybridised carbon atoms with a C-C bond length of 1.41 Å which is close to the

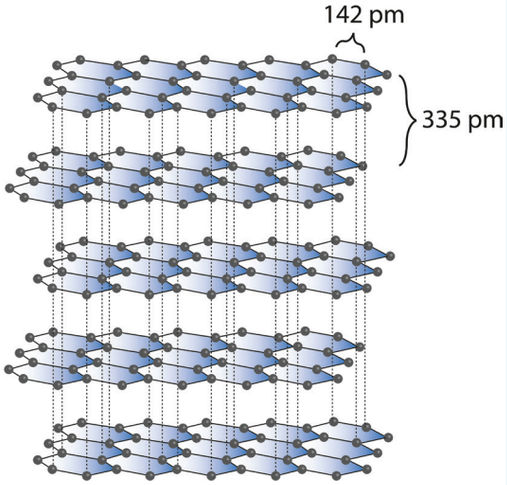


Figure 2.4 Structure of graphite

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